THE Z-GRADED TWIST ALGEBRA AND UNIVERSAL PSI ON RELATIVE PROJ

THE STACKS PROJECT AUTHORS

Editorial note. Closing 'Some details omitted.' covers the final assembly of the characterization; the construction is fully routed through the glueing engine with all inputs named and the uniqueness mechanism identified.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3760 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Relative glueing of sheaves. lemma-relative-glueing-sheaves (same chapter, judged non-self-contained this batch, not carded). Engine that glues the graded twist algebras over the basis of affine opens.
- Transition data over nested affines. lemma-proj-inclusion (carded this batch). Provides the transition isomorphisms between the twist algebras of the charts.
- Transitivity of transitions. lemma-transitive-proj (same chapter, bookkeeping citation, not carded). Verifies condition (d) of the glueing lemma on triples of affines.
- Functoriality of relative glueing. remark-relative-glueing-functorial (same chapter, not carded). Produces psi from the section-level maps A to the sections of the twists over Proj(A).

Source reference. Original equation 01MP: sources/stacks-constructions/pinned-original-source.tex, source lines 1949–1949. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NQ: sources/stacks-constructions/pinned-original-source.tex, source lines 3650–3650. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MY: sources/stacks-constructions/pinned-original-source.tex, source lines 2271–2271. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01NO: sources/stacks-constructions/pinned-original-source.tex, source lines 3597–3597. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01LI: sources/stacks-constructions/pinned-original-source.tex, source lines 374–374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NP: sources/stacks-constructions/pinned-original-source.tex, source lines 3630–3630. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 01LK: sources/stacks-constructions/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01NN: sources/stacks-constructions/pinned-original-source.tex, source lines 3580–3580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01NN. The morphism $\pi : \underline{Proj}_S(\mathcal{A}) \to S$ of Lemma 01NQ comes with the following additional structure. There exists a quasi-coherent $\mathbf{Z}$-graded sheaf of $\mathcal{O}_{\underline{Proj}_S(\mathcal{A})}$-algebras $\bigoplus_{n\in\mathbf{Z}} \mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(n)$, and a morphism of graded $\mathcal{O}_S$-algebras*

$$\psi : \mathcal{A} \longrightarrow \bigoplus\nolimits_{n\geq 0} \pi_* \left(\mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(n)\right)$$

uniquely determined by the following property: For every affine open $U \subset S$ with $A = \mathcal{A}(U)$ there is an isomorphism

$$\theta_U : i_U^* \left(\bigoplus\nolimits_{n\in\mathbf{Z}} \mathcal{O}_{Proj(A)}(n)\right) \longrightarrow \left(\bigoplus\nolimits_{n\in\mathbf{Z}} \mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(n)\right)|_{\pi^{-1}(U)}$$

of $\mathbf{Z}$-graded $\mathcal{O}_{\pi^{-1}(U)}$-algebras such that

$$\begin{array}{ccc} A_n & \xrightarrow{\qquad\qquad\psi\qquad\qquad} & \Gamma(\pi^{-1}(U), \mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(n)) \\ & {\scriptstyle (01MP)}\searrow \qquad \nearrow{\scriptstyle \theta_U} & \\ & \Gamma(Proj(A), \mathcal{O}_{Proj(A)}(n)) & \end{array}$$

is commutative.

Proof. We are going to use Lemma 01LI to glue the sheaves of $\mathbf{Z}$-graded algebras $\bigoplus_{n\in\mathbf{Z}} \mathcal{O}_{\mathrm{Proj}(A)}(n)$ for $A = \mathcal{A}(U)$, $U \subset S$ affine open over the scheme $\underline{\mathrm{Proj}}_S(\mathcal{A})$. We have constructed the data necessary for this in Lemma 01NO and we have

checked condition (d) of Lemma 01LI in Lemma 01NP. Hence we get the sheaf of $\mathbf{Z}$-graded $\mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}$-algebras $\bigoplus_{n\in\mathbf{Z}} \mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}(n)$ together with the isomorphisms θ_U for all $U \subset S$ affine open and all $n \in \mathbf{Z}$. For every affine open $U \subset S$ with $A = \mathcal{A}(U)$ we have a map $A \to \Gamma(\mathrm{Proj}(A), \bigoplus_{n\geq 0} \mathcal{O}_{\mathrm{Proj}(A)}(n))$. Hence the map ψ exists by functoriality of relative glueing, see Remark 01LK. The diagram of the lemma commutes by construction. This characterizes the sheaf of $\mathbf{Z}$-graded $\mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}$-algebras $\bigoplus \mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}(n)$ because the proof of Lemma 01MY shows that having these diagrams commute uniquely determines the maps θ_U. Some details omitted. □